Extraction of Symmetrical Components in the $\alpha\beta$ Frame

Technical Derivation

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1 Extraction of Positive and Negative Sequences in the Natural abc Reference Frame

1.1 The Complex Operator a

To handle the 120° displacement between phases, we define the complex operator a:

$$a = e^{j\frac{2\pi}{3}} = \cos\left(\frac{2\pi}{3}\right) + j\sin\left(\frac{2\pi}{3}\right) = -\frac{1}{2} + j\frac{\sqrt{3}}{2} \tag{1}$$

Useful properties of a include $a^3 = 1$ and $1 + a + a^2 = 0$.

1.2 Fundamental Equations (Fortescue)

Any unbalanced set of three-phase quantities $\mathbf{v}_{abc}$ can be expressed as:

$$v_a = v_0 + v_1 + v_2 \tag{2}$$

$$v_b = v_0 + a^2 v_1 + a v_2 \tag{3}$$

$$v_c = v_0 + a v_1 + a^2 v_2 \tag{4}$$

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} v_0 \\ v_1 \\ v_2 \end{bmatrix} \tag{5}$$

1.3 Derivation of the Components

1. Zero Sequence (v_0): Summing the three equations directly:

$$v_a + v_b + v_c = 3v_0 + v_1(1 + a^2 + a) + v_2(1 + a + a^2) \tag{6}$$

Since $(1 + a + a^2) = 0$, we get:

$$\mathbf{v_0} = \mathbf{\frac{1}{3}}(\mathbf{v_a} + \mathbf{v_b} + \mathbf{v_c}) \tag{7}$$

2. Positive Sequence (v_1): Multiply v_b by a and v_c by a^2, then sum:

$$v_a + av_b + a^2 v_c = (v_0 + v_1 + v_2) + a(v_0 + a^2 v_1 + av_2) + a^2(v_0 + av_1 + a^2 v_2) \tag{8}$$

$$= v_0(1 + a + a^2) + v_1(1 + a^3 + a^3) + v_2(1 + a^2 + a^4) \tag{9}$$

Using $a^3 = 1$ and $a^4 = a$:

$$\mathbf{v_1} = \mathbf{\frac{1}{3}}(\mathbf{v_a} + \mathbf{a v_b} + \mathbf{a^2 v_c}) \tag{10}$$

3. Negative Sequence (v_2): Multiply v_b by a^2 and v_c by a, then sum:

$$\mathbf{v_2} = \mathbf{\frac{1}{3}}(\mathbf{v_a} + \mathbf{a^2 v_b} + \mathbf{a v_c}) \tag{11}$$

1.4 Matrix Representation

The inverse Fortescue transform is given by:

$$\begin{bmatrix} v_0 \\ v_1 \\ v_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} \tag{12}$$

2 Extraction of Positive and Negative Sequences in the Natural abc Reference Frame

In a stationary reference frame, an unbalanced three-phase system (without zero sequence) is represented as a complex vector $\underline{v}_{\alpha\beta}$ which is the sum of a positive sequence component (rotating at $+\omega$) and a negative sequence component (rotating at $-\omega$).

2.1 Fundamental Equations

The complex space vector is defined as:

$$\underline{v}_{\alpha\beta}(t) = \underline{v}^+_{\alpha\beta}(t) + \underline{v}^-_{\alpha\beta}(t) \tag{13}$$

Using Euler's identity, we define the sequences as:

- **Positive Sequence:** $\underline{v}^+_{\alpha\beta}(t) = V^+ e^{j\omega t} = V^+(\cos\omega t + j \sin\omega t)$
- **Negative Sequence:** $\underline{v}^-_{\alpha\beta}(t) = V^- e^{-j\omega t} = V^-(\cos\omega t - j \sin\omega t)$

Separating the real (α) and imaginary (β) parts:

$$v_\alpha = V^+ \cos\omega t + V^- \cos\omega t \tag{14}$$

$$v_\beta = V^+ \sin\omega t - V^- \sin\omega t \tag{15}$$

We introduce the 90° phase-shift operator q, orthogonal transformation q, where $q = e^{-j\pi/2}$. This yields:

$$qv_\alpha = V^+ \sin\omega t + V^- \sin\omega t \tag{16}$$

$$qv_\beta = -V^+ \cos\omega t + V^- \cos\omega t \tag{17}$$

By algebraically combining the original and shifted signals, we extract the sequence components:
Positive Sequence Components

$$v^+_\alpha = \frac{1}{2}(v_\alpha - qv_\beta) \tag{18}$$

$$v^+_\beta = \frac{1}{2}(v_\beta + qv_\alpha) \tag{19}$$

Negative Sequence Components

$$v^-_\alpha = \frac{1}{2}(v_\alpha + qv_\beta) \tag{20}$$

$$v^-_\beta = \frac{1}{2}(v_\beta - qv_\alpha) \tag{21}$$

Matrix Implementation The extraction can be represented in matrix form for digital implementation:

$$\begin{bmatrix} v^+_\alpha \\ v^+_\beta \\ v^-_\alpha \\ v^-_\beta \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -q \\ q & 1 \\ 1 & q \\ -q & 1 \end{bmatrix} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} \tag{22}$$